Iuliia Kozina (Kewitz)

Applied Researcher | Medical Knowledge Representation & Information Retrieval

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RESEARCH PROFILE

Applied researcher with an MSc in Applied Mathematics and Informatics, an engineering background in biotechnology, an MBA, and 15+ years of experience working with complex knowledge-intensive systems. My current research focuses on how heterogeneous medical information can be transformed into structured, traceable, and expert-inspectable knowledge for use in medical AI and decision-support systems. My background combines biomedical NLP and knowledge graph research with extensive experience translating ambiguous real-world problems into analytical models, decision logic, and system requirements.

RESEARCH EXPERIENCE

MSc Thesis - Knowledge Graph Construction System for the Medical Field

Higher School of Economics | 2024-2026 | Moscow, Russia

Built and evaluated an end-to-end pipeline across 87 NCCN oncology clinical guidelines, producing a source-linked medical KG with 105K+ nodes and 143K+ semantic relations. The project combined biomedical NLP, terminology alignment, LLM-assisted relation extraction, and Graph RAG, while exposing the reliability constraints that must be addressed before such representations can be used in clinically meaningful decision-support workflows.

Founder & Research Lead

Gnodus | Jul 2026-Present | Germany | Early-stage initiative

• Developing a research agenda on transforming heterogeneous medical information into structured, traceable, and expert-inspectable knowledge assets.

• Mapping the research and implementation constraints of evidence provenance, semantic consistency, data access, and clinical validation.

SELECTED APPLIED SYSTEMS EXPERIENCE

Senior Strategy & Transformation Consultant

Deutsche Bahn Engineering & Consulting | Jun 2023-Present | Germany | Saudi Arabia | Kazakhstan

• Develop decision-support and analytical concepts for large-scale infrastructure programmes by translating operational, regulatory, financial, and data inputs into explicit models, assumptions, and analytical frameworks.

• Currently designing a national-level decision-support system (DSS) for passenger rolling-stock planning, linking operational KPIs, data structures, and decision logic into a consistent planning framework.

• Define functional and technical requirements for the DSS, translating policy, operational and planning needs into system logic, data requirements, and decision rules.

Co-founder | AI Product Strategy & Decision Support Systems Advisor

TradeDocAI | Jul 2025-Dec 2025 | Singapore

• Translated international trade standards and document workflows into structured data requirements, verification logic, and product specifications, supporting use-case prioritisation and high-level solution design for an AI-enabled trade-document platform.

• Formalised standards mappings, cross-document validation rules, and exception paths to make document checks more interoperable, auditable, and consistent across heterogeneous trade-document workflows.

RESEARCH INTERESTS

Medical knowledge representation and integration: semantic alignment, context preservation, provenance, and evidence across heterogeneous medical sources.

Medical information retrieval and evidence grounding: connecting AI systems to current, source-linked, and clinically relevant evidence.

Reasoning and decision support: methods for making intermediate inferences, assumptions, and resulting claims available for inspection and verification.

TECHNICAL ENVIRONMENT & RESEARCH METHODS

Clinical NLP & knowledge representation: FAISS · UMLS and biomedical terminologies · Neo4j · Cypher

Language models, retrieval & structured prediction: PyTorch · Hugging Face Transformers · LangChain · Graph RAG · NER and relation extraction

Applied ML & evaluation: Python · Pandas · NumPy · Scikit-learn · validation design · comparative model evaluation · error analysis

Reproducible prototypes & workflows: Git/GitHub · MLflow · Apache Airflow

Research practice: Literature synthesis and evidence mapping · structured literature search and screening in PubMed, Scopus and Semantic Scholar · citation management in Zotero.

EDUCATION

MSc, Applied Mathematics and Informatics (Diploma with honours)

Higher School of Economics, Russia | 2024-2026

MBA

National University of Singapore, Singapore | 2019-2021

MSc, Innovative Technologies

Saint-Petersburg State University of Aerospace Instrumentation, Russia | 2009-2011

MEng, Biotechnology

Saint-Petersburg State Technological Institute, Russia | 2001-2007

SELECTED TRAININGS & CERTIFICATES

Machine Learning Specialisation

Stanford Online / DeepLearning.AI (2026)

Deep Learning Engineer (Neural Networks): Natural Language Processing / Data Science

Yandex Ltd. (2025-2026)

PROFESSIONAL EXPERIENCE: Complex Systems, Knowledge & Decision Support

My work spans transport, infrastructure, logistics and health-system analytics. Across these settings, the recurring task was to convert fragmented data, documents, standards and stakeholder knowledge into explicit models, decision logic, and implementation-ready structures. This experience informs my current research interest in traceable knowledge representations: the challenge is not simply extracting information, but preserving its context, assumptions, and accountability as it moves between experts, systems, and decisions.

Business Management Lead | Business Transformation & Knowledge Management

Deutsche Bahn Engineering & Consulting | May 2021-Jun 2023 | Central Asia | Germany

- Led analytical and strategic work in national railway transformation programmes, connecting strategy, operating models, data structures, and knowledge-management practices across organisational boundaries.
- Designed long-term roadmaps, KPI frameworks, and initiative portfolios; made assumptions, dependencies, and decision criteria visible across business and government stakeholders.

Market Intelligence Analyst | Healthcare Analytics

Ministry of Health | Dec 2019-Mar 2020 | Singapore

- Analysed operational performance and patient-experience information across Singapore's public healthcare system to support evidence-informed service improvement recommendations.

Senior Transformation Manager | Strategy Manager | Analyst

Russian Railways | Jan 2010-Jun 2019 | Russia | Finland

- Worked across performance management, process architecture, predictive and operational modelling, customer analytics and strategic transformation in a large operational organisation. Built KPI systems, analytical frameworks, and decision-support structures used by cross-functional teams and senior stakeholders.

SELECTED CROSS-DOMAIN CAPABILITIES

- Frame ambiguous operational or policy problems as explicit questions, requirements, assumptions, and boundaries.
- Integrate fragmented data, documents, standards, and expert inputs into shared analytical or decision models.
- Make rules, constraints, dependencies, and performance measures explicit and open to review.
- Design decision-support and validation workflows subject to operational, governance, and implementation constraints.
- Work with domain experts and cross-functional stakeholders to test, challenge, and refine model assumptions.